

Pablo Villacorta Aylagas

Telecommunications Engineer
PhD Candidate

pablo.villacorta@uva.es

May 29, 2020

Valladolid, Spain

About me

I am a Telecommunications Engineer interested in all its areas, from signal processing to programming and telematics, including disciplines such as electronics and instrumentation. I am also passionate about Magnetic Resonance Imaging (MRI), a field that brings together the best of each discipline and offers room for innovation to improve people's health.

Currently, I am working as a researcher in the Image Processing Lab (LPI) at Universidad de Valladolid (Spain), where I contribute to the **KomaMRI** project while developing a **web-based tool for MRI pulse sequence editing and simulation**.

Resume

Experience

MRI Researcher – PhD Candidate

Image Processing Lab (LPI) – Universidad de Valladolid

2022 – Present

Working on MRI simulation frameworks, with a focus on modeling complex phenomena like blood flow and non-rigid motion. Also involved in optimizing simulation pipelines and improving algorithms performance and accuracy.

Education

Universidad de Valladolid, Spain

MSc in Telecommunications Engineering

2022 – 2023

Habilitating MSc focused on real-world projects. We designed an IoT datalogger with LoRa and sensor integration, and deployed a 4G-based system for real-time video processing and AI traffic sign detection.

Universidad de Valladolid, Spain

BSc in Telecommunications Engineering

2018 – 2022

Acquired a solid foundation in signal processing, programming, electronics, systems engineering, and telematics.

Languages

English – C1 Level

Certified with Cambridge Certificate in Advanced English (CAE)

2026

Score: 198

Certified with Cambridge First Certificate in English (FCE)

2020

Score: 182

Publications

Journal Papers

2026

A multi-*b*-value test-retest diffusion MRI brain dataset for model validation and reproducibility assessment

Sumbitted to Scientific Data. (Preprint) DOI: [10.64898/2026.08.23.746449](https://doi.org/10.64898/2026.08.23.746449)

T. Pieciak, I. Guadilla, D. Ciupek, R. Navarro-González, S. Merino-Caviedes, **P. Villacorta-Aylagas**, L. Magdaleno Humayor, M. Villa Aparicio, J. Rueda-Ramos, R. Santiesteban-Mendo, R. Moro-Boyero, A. Tristán-Vega

MRSeqStudio: MRI Sequence Design and Simulation as a Service in a Free and Open-Source Web Platform

J Med Syst, 50, 65 (2026). DOI: [10.1007/s10916-026-02394-1](https://doi.org/10.1007/s10916-026-02394-1)

P. Villacorta-Aylagas, M. Rodríguez-Cayetano, C. Castillo-Passi, P. Irarrazaval, F. Simmross-Wattenberg, C. Alberola-López

Versatile and Highly Efficient MRI Simulation of Arbitrary Motion in KomaMRI

Magn Reson Med, 95, no. 3 (2026): 1791-1803. DOI: [10.1002/mrm.70145](https://doi.org/10.1002/mrm.70145)

P. Villacorta-Aylagas, C. Castillo-Passi, R. Kierulf, R.M. Menchón-Lara, J.R. Rodríguez-Galván, J. Sierra-Pallares, P. Irarrazaval, C. Alberola-López

Sampling of non-Gaussian Ensemble Average Propagators for the simulation of diffusion magnetic resonance images

Magn Reson Med, 95, no. 2 (2026): 1255-1265. DOI: [10.1002/mrm.70080](https://doi.org/10.1002/mrm.70080)

J.R. Rodríguez-Galván, **P. Villacorta-Aylagas**, S. Merino-Caviedes, F. Simmross-Wattenberg, C. Castillo-Passi, P. Irarrazaval, A. Tristán-Vega, C. Alberola-López

Conference Contributions

2026

Enhancing KomaMRI with a Pulseq I/O Interface for Reproducible Sequence Workflows

2026 ISMRM Iberian Chapter Meeting, Coimbra, Portugal

P. Villacorta-Aylagas, C. Castillo-Passi, P. Irarrazaval, F. Simmross-Wattenberg, M. Rodríguez-Cayetano, C. Alberola-López

MRSeqStudio: A Free and Open-Source Web Application for MR Sequence Design and Simulation as a Service

2026 ISMRM & ISMRT Annual Meeting & Exhibition, Cape Town, South Africa

P. Villacorta-Aylagas, C. Castillo-Passi, P. Irarrazaval, F. Simmross-Wattenberg, M. Rodríguez-Cayetano, C. Alberola-López

2025

A Free and Open-Source Web Application for Pulse Sequence Development and Simulation

1st Best Poster Presentation

2025 ISMRM Iberian Chapter Meeting, Barcelona, Spain

P. Villacorta-Aylagas, C. Castillo-Passi, P. Irarrazaval, F. Simmross-Wattenberg, M. Rodríguez-Cayetano, C. Alberola-López

2024

Flexible Motion Models for GPU-accelerated MRI Simulations with Complex Motion and Flow using KomaMRI

ESMRMB 2024, Barcelona, Spain

P. Villacorta-Aylagas, C. Castillo-Passi, R.M. Menchón-Lara, J. Anatol-Hernández, P. Irarrazaval, J. Sierra-Pallares, C. Alberola-López

Proof of Concept for the Simulation of Diffusion Models in the Myocardium through KomaMRI

CASEIB 2024, Sevilla, Spain

J. R. Rodríguez-Galván, **P. Villacorta-Aylagas**, S. Merino-Caviedes, C. Castillo, J. Sánchez-González, D. Filgueiras-Rama, A. Tristán-Vega, P. Irarrazaval, C. Alberola-López

New Methods for Defining Dynamic Phantoms in KomaMRI

2024 ISMRM Iberian Chapter Meeting, Porto, Portugal

P. Villacorta-Aylagas, C. Castillo-Passi, R.M. Menchón-Lara, J. Anatol-Hernández, P. Irarrazaval, J. Sierra-Pallares, C. Alberola-López

Synthetic spin trajectory generation for advanced Flow phantom in MRI simulation

2024 ISMRM Iberian Chapter Meeting, Porto, Portugal

J. Anatol-Hernández, J. Sierra-Pallares, **P. Villacorta-Aylagas**, C. Castillo-Passi, R.M. Menchón-Lara, P. Irarrazaval, C. Alberola-López

A KomaMRI Phantom Extension for Integrated Dynamic Imaging

2024 ISMRM & ISMRT Annual Meeting & Exhibition, Singapore

P. Villacorta-Aylagas, C. Castillo-Passi, R. M. Menchón-Lara, P. Irarrazaval, J. B. Sierra-Pallares, C. Alberola-López

2023

A Web Version of KomaMRI. Sequence Editor and Remote Execution

2023 ISMRM Iberian Chapter Meeting, Valladolid, Spain

P. Villacorta-Aylagas, I. Fernández-Arias, C. Castillo-Passi, P. Irarrazaval, F. Simmross-Wattenberg, M. Rodríguez Cayetano, C. Alberola-López

Parallel Device-Agnostic Simulation of Magnetic Resonance Imaging Sequences

2023 ISMRM Iberian Chapter Meeting, Valladolid, Spain

I. Fernández-Arias, **P. Villacorta-Aylagas**, C. Castillo-Passi, P. Irarrazaval, F. Simmross-Wattenberg, M. Rodríguez-Cayetano, C. Alberola-López

2022

A Cross-Plattform Editor and Simulator of Magnetic Resonance Imaging Sequences. Design and Implementation

CASEIB 2022, Valladolid, Spain

P. Villacorta-Aylagas, I. Fernández-Arias, C. Castillo-Passi, P. Irarrazaval, F. Simmross-Wattenberg, M. Rodríguez Cayetano, C. Alberola-López

Simulación paralela multidispositivo de secuencias de imagen por resonancia magnética

CASEIB 2022, Valladolid, Spain

I. Fernández-Arias, **P. Villacorta-Aylagas**, C. Castillo-Passi, P. Irarrazaval, F. Simmross-Wattenberg, M. Rodríguez-Cayetano, C. Alberola-López